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Milestone 1: Data Representation & Foundations

1. Dataset Overview

This dataset was downloaded from Kaggle as a CSV file (smmh.csv). It looks at the relationship between social media habits and mental health outcomes, based on responses from 481 people who filled in a survey between April and November 2022.

What made this dataset interesting to work with is that it goes deeper than the usual engagement metrics. Rather than just tracking likes or follower counts, it captures how people actually feel when using these platforms. That includes things like anxiety levels, sleep quality, and how often someone finds themselves comparing their life to what they see online. Those kinds of indicators tend to get overlooked in purely numbers-based datasets, so having them here gave us a lot more to explore.

Each row represents one person who filled in the survey. The dataset has **481 responses**, collected between April and November 2022.

Some of the main things recorded include:

- Age and gender
- Relationship status and occupation
- Platforms used (like YouTube, Instagram, etc.)
- Time spent on social media
- Mental health indicators (rated from 1 to 5)

The dataset has **21 columns**, which we grouped into:

- Demographics
- Usage behaviour
- Mental health scales
- Outcome indicators

One thing to note is that the mental health values are based on a **scale (1-5)**, where:

Score 1	Very Low / Never: the respondent rarely or never experiences this
Score 2	Low / Rarely: infrequent occurrence
Score 3	Moderate / Sometimes: occurs occasionally
Score 4	High / Often: occurs frequently
Score 5	Very High / Always: the respondent almost always experiences this

This scale is used consistently across all the mental health columns, so a higher number always means a more intense or more frequent experience of that indicator.

2. Data Quality Assessment

Before doing any analysis, we checked if the data was actually clean and usable. We looked at four main things:

1. Completeness

We used `df.isnull().sum()` to scan every column for null (missing) values. Most columns came back completely clean. The only exception was the Affiliations column, which had about 30 missing entries out of 481.

Since Affiliations is not one of the core variables we were planning to analyse (the mental health scales and usage time were all 100% complete), we decided not to drop those rows. Instead, we filled the gaps with the string 'N/A' using `df['Affiliations'].fillna('N/A')`. That way the column looks complete and does not throw errors later on when we are doing things like grouping or counting.

2. Uniqueness

We ran `df[df.duplicated()]` to check for any rows that were exact copies of each other. There were no genuine duplicates in the dataset. The one thing worth flagging is that out of 481 rows, only 480 timestamps were unique. This means two responses came in at the exact same second. After looking at the actual data for those rows, the responses themselves were different, so it is more likely just a coincidence than an actual double submission. No rows were removed.

3. Consistency

We printed `df.dtypes` to confirm that Python had assigned the right data types to each column. Numbers needed to be stored as numbers, and text needed to be stored as objects (strings). Everything looked correct:

- The mental health Likert scale columns (Worry, Depression, Sleep Issues, etc.) were stored as `Int64`, which is correct since the values are whole numbers between 1 and 5.
- Age was stored as `Float64`.
- Categorical fields like Gender, Relationship Status, and Platforms Used were stored as `Object` (string), which is what we expected.
- The Time Spent column was also stored as an `Object` because it uses text ranges like 'Between 2 and 3 hours' rather than raw numbers. This is actually fine, since those categories are not something we can average directly anyway.

No type conversions were needed.

4. Validity

We also checked that the values in each column made sense. For the mental health scales, all values were confirmed to be between 1 and 5. For the Time Spent column, the unique values were printed using `df['Time_Spent'].unique()` to confirm that only valid survey categories appeared. Because the original survey used a fixed multiple-choice format, there was no way for someone to enter a nonsense value like '47 hours.' No issues were found here

3. Outlier Analysis

To check for outliers, we used the IQR (Interquartile Range) method. The basic idea is that you find the middle 50% of the data (the range between the 25th and 75th percentiles), then flag anything that falls too far outside that range.

The formula for the boundaries is:

- Lower Bound: $Q1 - (1.5 \times IQR)$
- Upper Bound: $Q3 + (1.5 \times IQR)$

We applied this to all the numeric columns in the dataset. Anything outside those bounds gets flagged as a potential outlier. We used IQR specifically because it works well even when distributions are skewed, which ours clearly are (most respondents are young, and several mental health variables have scores that cluster toward one end of the scale).

3.2 What We Found

Column	Outliers Found	Direction	Why This Happens
Age	84	High end	Most respondents are ~21. Anyone older than ~34 gets flagged, even though older people using social media is completely normal.
Restlessness	44	Low end	Most people scored in the mid-to-high range, so the few who scored very low (1 or 2) stand out statistically.
Comparison Feelings	33	Low end	Same cluster effect: the majority scored
Distractibility Scale	33	Low end	Narrow central distribution makes low-end values look extreme.
Purposeless Usage	20	Low end	Similar pattern: most people admitted to s are uncommon.
All other columns	0	:	Responses were spread well enough across 1–5 that no extreme values were flagged.

We decided to keep all the outliers in the dataset rather than removing them. This came down to one main reason: these are real survey responses from real people. The 84 'age outliers' are not errors, they are just older participants who happen to be in the minority compared to the mostly university-aged respondents. Removing them would mean deliberately excluding people based on their age, which would introduce a bias that makes the dataset less representative.

The same logic applies to the mental health scale outliers. Someone scoring a 1 on the Restlessness scale is just a person who does not feel restless without their phone. That is a valid response, not a data error.

4. Descriptive Statistics

Once the data was confirmed to be clean, we calculated basic summary statistics across the key numeric variables. This gives a general picture of what the 'typical' respondent looks like and how spread out the responses are.

4.1 Summary Statistics Table

Variable	Mean Mode Std Dev	Key Observation
Age	26.14 21 9.92	Right-skewed: most respondents older outliers pull the mean up to 26.
Purposeless Usage	3.55 4 1.10	Fairly high: the typical social media without a clear purpose.
Worry Scale	3.56 5 1.28	Left-skewed: the most common of 5. Elevated worry is very common here.
Depression Frequency	3.26 3 1.31	Near-symmetric: responses are scale. The median and mean are close.
Sleep Issues	3.20 5 1.46	High variance (2.14): sleep differently. The mode is 5 but the mean is 3.2.
Social Comparison	2.83 1 1.41	Right-skewed: this is one of the common answer was 1 (never), yet the average is still close to 3.

4.2 Distribution Shapes

Looking at the shape of each distribution helped us understand the data beyond just the averages:

Age	Right-skewed. The bulk of responses are packed around 21, with a long tail stretching toward the 91-year-old maximum. This confirms the sample is dominated by younger adults.
Worry Scale	Left-skewed. Scores heavily cluster at 4 and 5, meaning high worry is the norm in this sample, not the exception.
Depression Frequency	Roughly symmetric. Mean (3.26) and median (3.0) are close together, which means responses are spread fairly evenly without a strong pull in either direction.
Sleep Issues	Wide spread with a high variance of 2.14. The mode is 5 but the mean is 3.2, meaning some people are fine while a significant portion is really struggling with sleep.
Likert scales generally	Platykurtic (flat). Because people spread their responses across all five options, the distribution is flatter than a normal bell curve. There is no single 'most typical' experience.

One of the more interesting findings from the descriptive analysis is that the Time Spent variable is bimodal meaning, the data has two separate peaks rather than clustering around one central value. This suggests the sample is not one uniform group but is actually made up of two distinct types of users:

Casual users (**between 2-3hrs**) and heavy users (**5+hrs**)

5. Platform Usage Analysis

The Platforms column gave us a bit of a headache. Unlike most columns where each cell has a single value, this one stores a list of platforms per person, something like 'Facebook, Instagram, YouTube' all in one cell.

If we tried to count platforms directly using something like `value_counts()`, pandas would treat 'Facebook, Instagram' as one unique value, completely different from just 'Facebook'. That would give completely wrong results. To fix this, we used the `.explode()` function in pandas after splitting the comma-separated strings into lists. This turns each platform in a cell into its own row, which then lets us count each platform independently and accurately.

5.1 Platform Popularity

Platform	Approximate User Count	Notes
YouTube	400+	Highest usage across all platforms
Facebook	400+	Essentially tied with YouTube
Instagram	~350	Third most used
Twitter / X	100-200	Moderate presence
Discord	100-200	Moderate presence
Reddit	100-200	Moderate presence
Snapchat	100-200	Moderate presence
TikTok	< 100	Lowest in this dataset likely because the data is from 2022 when TikTok had not yet peaked

After exploding the Platforms column, another clear pattern emerged: the vast majority of respondents are not sticking to just one platform. Most people listed three, four, or even more platforms in their response.

6. Correlation Analysis

We built a Pearson correlation matrix covering four of the main numeric variables: Age, Worry Scale, Depression Frequency, and Sleep Issues.

6.2 Correlation Matrix Values

	Age	Worry_Scale	Depression_freq	Sleep_Issues
Age	1.00	-0.29	-0.30	-0.16
Worry_Scale	-0.29	1.00	0.59	0.33
Depression_freq	-0.30	0.59	1.00	0.37
Sleep_Issues	-0.16	0.33	0.37	1.00

Worry, Depression, and Sleep Issues are all positively correlated with each other. The strongest relationship is between Worry and Depression ($r = 0.58$), which makes intuitive sense. If someone is frequently anxious, they are likely also experiencing depressive episodes more often, and vice versa.

Worry and Sleep Issues also show a meaningful positive relationship ($r = 0.47$), as do Depression and Sleep Issues ($r = 0.55$). This suggests that poor mental health and poor sleep tend to go hand in hand in this dataset.

Age has a weak negative correlation with all three mental health variables (around -0.05 to -0.08). Older respondents in the sample tend to score slightly lower on worry, depression, and sleep issues - but the relationship is so weak that it would be a stretch to draw any strong conclusions from it.

7. Issues Encountered

While working with this survey data, we ran into a few problems that made the cleaning process take a lot longer than expected.

1. Messy Column Headers

The biggest issue was that the column names in the CSV were actually the full survey questions. For example, instead of just saying "Age," the column was named "1. What is your age?" This made the code look incredibly messy and it difficult to type out and we had to rename every single column to a shorter version before starting.

2. Time Spent was not Numerical

The "Time Spent" column was actually stored as text ranges like "Between 2 and 3 hours" or "More than 5 hours." This meant we couldn't just calculate an average or a sum immediately. We had to treat it as a categorical variable instead of a continuous one, which changed how we built some of the visualization.

3. Multiple Values in One Cell

The Platforms column was probably the messiest one to work with. Each person's response was a comma-separated list of apps inside a single cell, so one cell might say 'Facebook, Instagram, TikTok'. If we tried to count those normally with `value_counts()`, pandas would read the entire string as one unique entry. That meant 'Facebook, Instagram' and 'Facebook, Instagram, Twitter' would both count as completely separate categories.

To fix this, we had to split each cell into a list and then use `.explode()` to break each platform out into its own row. Only after that could we count them properly and see which apps were actually the most popular.

4. Missing Affiliations

About 30 people didn't answer the question about what organizations they belong to. While it didn't break the whole dataset, it left "NaN" values everywhere in that column. We decided to just label them as "N/A" so the data looked complete and didn't throw errors later on.

8. Summary and Conclusions

Here is a quick rundown of the main takeaways from this milestone:

- The dataset was largely clean before analysis. The only fix needed was handling the 30 missing Affiliations entries. No rows were removed.
- The sample skews heavily young - the modal age is 21 and the median is 22, which means a large chunk of respondents are university students. The mean of 26 is pulled up by the smaller number of older participants. Since the sample is overwhelmingly made up of young adults findings probably do not generalize well to older populations.
- Social media use is high overall, and the time distribution is bimodal. There are two groups: casual users (2-3 hrs/day) and heavy users (5+ hrs/day), with the heavy-user group being the larger of the two.
- YouTube and Facebook are the most widely used platforms (both at 400+ users), followed by Instagram. Most respondents use multiple platforms, not just one.
- Worry and sleep issues stand out the most. Both have a mode of 5 out of 5, and worry, depression, and sleep problems are all positively correlated with each other.

Milestone 2: Data Pipeline

1. Project overview

This notebook implements a four-stage ETL (Extract-Transform-Load) pipeline that processes a raw social media survey dataset (smmh.csv) to produce cleaned, feature-enriched, and aggregated outputs ready for downstream analysis and visualisation.

aggregation, and final output generation: all with full reproducibility via a JSON audit trail.

1.1 Dataset Summary

Attribute	Value
Source file	smmh.csv
Initial rows	481
Initial columns	21
Rows after cleaning	478
Final columns (after features)	33
Survey period (sample)	April 2022
Population	Social media users (Q6 = Yes)

1.2 Pipeline Architecture

The pipeline is structured as four sequential stages, each saving its outputs to a dedicated directory:

- Stage 1: Ingest & Clean → data/02_clean/
- Stage 2: Feature Engineering→ data/03_features/
- Stage 3: Transform & Aggregate → data/04_transformed/
- Stage 4: Output & Report → data/05_output/

2. Column Schema & Renaming

All 20 raw survey question headers are renamed to concise, underscore-separated aliases via COL_MAP. The table below lists every mapping and the resulting data type after cleaning.

Original Question	Alias	Type	Description
1. What is your age?	Age	int	Respondent age (years)
2. Gender	Gender	string	Male / Female / Non-binary/Other
3. Relationship Status	Relationship_Status	string	Single / Married / In a relationship / Divorced
4. Occupation Status	Occupation_Status	string	University Student / Salaried Worker / School Student / Retired
5. Organization type	Organization	string	University / Company / School / Government / NGO
6. Do you use social media?	Uses_Social_Media	string	Yes / No (filter: only Yes retained)
7. Platforms used	Platforms	string	Comma-separated list of platforms
8. Average daily time	Daily_Usage → Daily_Minutes	string → int	Parsed to midpoint minutes (see §3.2)
9. Purposeless usage	Purposeless_Usage	int 1–5	Likert: compulsive / aimless use
10. Distraction while busy	Distraction_While_Busy	int 1–5	Likert: distracted during tasks
11. Restlessness	Restlessness	int 1–5	Likert: restless without social media
12. Easily distracted	Easily_Distracted	int 1–5	Likert: general distractibility
13. Bothered by worries	Bothered_By_Worries	int 1–5	Likert: anxiety / worry level
14. Concentration difficulty	Concentration_Difficulty	int 1–5	Likert: difficulty concentrating
15. Social comparison	Social_Comparison	int 1–5	Likert: comparing self to others
16. Comparison feelings	Comparison_Feelings	int 1–5	Likert: sentiment from comparisons
17. Validation seeking	Validation_Seeking	int 1–5	Likert: seeking likes/approval
18. Depression frequency	Depression_Frequency	int 1–5	Likert: frequency of low mood
19. Interest fluctuation	Interest_Fluctuation	int 1–5	Likert: varying interest in activities
20. Sleep issues	Sleep_Issues	int 1–5	Likert: frequency of sleep problems

3. Ingest & Clean

Purpose: Load the raw CSV, rename columns, parse categorical usage-time values to numeric minutes, standardise gender, remove non-users, impute sparse missing values, remove duplicates, and cap statistical anomalies.

3.1 Inputs & Outputs

Type	File / Path	Description
Input	smmh.csv	Raw survey CSV (481 rows × 21 columns)
Output	data/02_clean/social_media_clean.csv	Cleaned dataset (478 rows × 23 columns)
Output	data/02_clean/cleaning_report.json	Full audit trail with timestamps and counts

3.2 Daily Usage Parsing

The ordinal text categories in the Daily_Usage column are converted to numeric midpoint minutes by the parse_daily_minutes() helper function:

Survey Category	Midpoint (Minutes)
Less than an Hour	30
Between 1 and 2 hours	90
Between 2 and 3 hours	150
Between 3 and 4 hours	210
Between 4 and 5 hours	270
More than 5 hours	330

3.3 Cleaning Steps

Step	Method	Detail
Column renaming	df.rename(columns=COL_MAP)	All 20 survey questions renamed via COL_MAP dictionary
Daily usage parsing	parse_daily_minutes()	Text categories → numeric midpoints (see §3.2)
Gender standardisation	standardise_gender()	Free-text → Male / Female / Non-binary/Other
Non-user filter	df[Uses_Social_Media == 'yes']	Excludes respondents who do not use social media
Missing values: Age	Median imputation	NaN filled with median age
Missing values:	Median imputation	NaN filled with median minutes
Missing values:	Median imputation	0 missing values found; no action required
Null User_ID rows	df.dropna(subset=['User_ID'])	Rows with no User_ID are dropped
Duplicate removal	df.drop_duplicates()	Full-row deduplication (0 duplicates found)
Anomaly capping	3×IQR fence + clip()	Applied to Age, Daily_Minutes, and all 12 Likert cols

3.4 Anomaly Detection: 3×IQR Fence

For each numeric column the pipeline computes:

- Lower fence = $\max(0, Q1 - 3 \times IQR)$
- Upper fence = $Q3 + 3 \times IQR$
- Values outside the fence are clipped (not dropped) to preserve sample size.

Result: 55 Age values were clipped to the range [6.0, 41.0]. No Likert outliers were detected.

3.5 Actual Output (Stage 1)

Final shape after all cleaning steps: 478 rows × 23 columns.

4. Feature Engineering

Purpose: Derive composite mental-health scores, usage and risk tiers, platform count, and age group from the cleaned dataset. Adds 10 new columns to produce a feature-rich dataset.

4.1 Composite Mental-Health Scores

Five scores are computed as the row-wise mean of their constituent Likert items (rounded to 3 d.p.):

Feature	Source Likert Items	Captures
Addiction_Score	Purposeless_Usage, Distraction_While_Busy, Restlessness	Compulsive / aimless social media use
Attention_Score	Easily_Distracted, Bothered_By_Worries, Concentration_Difficulty	Attention deficit and anxiety
Social_Anxiety_Score	Social_Comparison, Comparison_Feelings, Validation_Seeking	Social comparison and approval seeking
Wellbeing_Score	Depression_Frequency, Interest_Fluctuation, Sleep_Issues	Depression, interest, and sleep quality
MH_Impact_Score	All 12 Likert columns (Q9–Q20)	Grand mean: overall mental-health

4.2 Additional Derived Features

Feature	Formula / Method	Purpose
MH_Impact_Pct_Rank	rank(pct=True) on MH_Impact_Score	Relative standing of each respondent (0–1 scale)
Platform_Count	len(Platforms.split(','))	Number of social media platforms used simultaneously
Usage_Tier	pd.cut on Daily_Minutes: 0–90 / 91–150 / 151–270 / 271–340 min	Categorical usage band: Low / Moderate / High / Very High
MH_Risk_Tier	pd.cut on MH_Impact_Score: 1–2 / 2–3 / 3–4 / 4–5	Categorical risk tier: Low / Moderate / High / Very High Risk
Age_Group	pd.cut on Age: ≤18 / 19–24 / 25–34 / 35+	Life-stage groupings: Teen / Young Adult / Adult / Older

4.2 Output

File: data/03_features/social_media_features.csv: 478 rows × 33 columns (original 23 + 10)

5. Transform & Aggregate

Purpose: Produce platform-level summaries, demographic cross-tabs, a usage-tier × risk-tier pivot, and a Pearson correlation matrix.

5.1 Platform Aggregation

The Platforms column (comma-separated) is split and exploded so each user appears once per platform they use. Only the 10 known platforms are retained. The resulting dataset is grouped by Platform_Single to produce per-platform statistics.

Sample output (sorted by Avg_Daily_Minutes descending):

Platform	Users	Avg Daily (min)	Avg MH Impact	Avg Addiction	Avg Attention	Share %
TikTok	94	245.1	3.501	3.546	3.830	4.59%
Snapchat	181	230.6	3.314	3.368	3.654	8.85%
Discord	197	222.8	3.198	3.223	3.555	9.63%
Pinterest	144	221.3	3.176	3.199	3.433	7.04%
Instagram	358	218.5	3.222	3.270	3.544	17.50%
YouTube	410	209.4	3.131	3.184	3.406	20.04%
Facebook	407	207.6	3.150	3.207	3.416	19.89%
Reddit	124	207.1	3.106	3.113	3.521	6.06%
Twitter	131	205.9	3.049	3.089	3.392	6.40%

5.2 Demographic Cross-Tabs

Three cross-tabulations are built from the feature dataset, each computing count, average daily minutes, average MH impact, average addiction score, and average wellbeing score:

- Gender summary: grouped by Gender (3 rows × 6 columns)
- Age group summary: grouped by Age_Group (4 rows × 6 columns)
- Occupation summary: grouped by Occupation_Status (4 rows × 4 columns)

5.3 Usage-Tier × MH-Risk Pivot

A pivot_table cross-tabulates Usage_Tier (rows) against MH_Risk_Tier (columns), with cell values representing user counts. This exposes the distribution of mental-health risk across usage intensity bands.

5.4 Correlation Matrix

A 19×19 Pearson correlation matrix is computed across: Daily_Minutes, Platform_Count, MH_Impact_Score, the four composite scores, and all 12 Likert items.

5.5 Output Files

File	Dimensions	Description
platform_summary.csv	9×10	Per-platform user count, avg usage, avg MH scores
gender_summary.csv	3×6	MH metrics grouped by gender
age_group_summary.csv	4×6	MH metrics grouped by age group
occupation_summary.csv	4×4	MH metrics grouped by occupation
usage_mh_risk_pivot.csv	4×5	Usage tier × MH risk cross-tab (user counts)
correlation_matrix.csv	19×19	Pearson correlation across all key numeric columns

6. Output & Report

Purpose: Consolidate the master enriched dataset and write a pipeline manifest capturing run metadata, stage list, and column inventory.

6.1 Output Files

File	Description
data/05_output/master_dataset.csv	Final respondent-level dataset: all original and engineered
data/05_output/pipeline_manifest.json	Run timestamp, ordered stage list, full column inventory

7. Key Design Decisions

Decision	Rationale
3×IQR fence (not 1.5×)	Survey data has natural heavy tails; 1.5× IQR would aggressively clip valid extreme responses.
Clip rather than drop outliers	Preserves sample size and avoids survivorship bias.
Median imputation for missing values	Robust to skewed distributions in age and usage data: preferred over
Grand-mean MH_Impact_Score across all 12 items	Simple, interpretable overall index covering all mental-health dimensions equally.
Platform explode strategy	Allows per-platform analysis while keeping every user's data intact.
Fixed bins for Usage_Tier	Midpoint-based breaks (90, 150, 270 min) align exactly with the survey's ordinal categories.
User_ID as 'U_' + (index+1)	Creates a stable, human-readable unique identifier for each respondent.
Only 10 known platforms retained	Strips noise from free-text responses; ensures only analysable categories enter the summary.

8. Runtime Environment & Dependencies

Library	Usage
pandas	Data loading, cleaning, grouping, pivot tables, CSV I/O
numpy	Quantile / IQR computation, NaN handling, numeric operations
json	Serialising the cleaning report and pipeline manifest
os	Directory creation (os.makedirs)
datetime	Run timestamp in audit reports
Python 3.10.0	Runtime version (Google Colab environment)

MILESTONE 3

Exploratory Data Analysis (ED)

1. Overview

Milestone 3 is where the project shifts from cleaning and preparing data to actually exploring it visually. After finishing the feature engineering in Milestone 2, we loaded the processed dataset and started building charts to see what stories the numbers were telling us.

The goal at this stage was not just to produce pretty graphs but to use visualisation as an analytical tool: statistical testing in Milestone 4.

1.1 Tools and Setup

We used three main libraries throughout this milestone:

- Matplotlib: for the core chart structure and layout control
- Seaborn: for higher-level statistical plots like KDE curves, violin plots, and heatmaps
- SciPy Stats: for regression lines and correlation values overlaid on the charts

We also loaded four pre-processed CSV files that were generated during Milestone 2: the main feature-engineered dataset (`social_media_features.csv`), a platform summary, a gender summary, and an age group summary. Having these pre-aggregated tables ready made the visualisation code a lot cleaner.

2. Charts and Visualisations

We produced nine charts across this milestone, each targeting a different aspect of the data. The table below gives a quick overview before we go into each one in detail:

Chart	Type	Variables	Purpose
1	Bar Chart	Platform vs Avg Daily Minutes	Show which platforms consume the most time
2	Line Chart	Age Group vs MH Impact & Wellbeing	Track how mental health changes across age groups
3	Scatter Plot	Daily Minutes vs Addiction Score (by MH Risk Tier)	Explore the relationship between usage time and addiction
4	Histogram + KDE	MH Impact Score (full population)	Show the distribution shape of the main outcome variable
5	KDE by Gender	MH Impact Score split by Male/Female	Compare distributions across gender groups
6	Correlation Heatmap	All Likert items + engineered scores	Map all variable relationships in one view
7	Grouped Bar Chart	MH Impact, Addiction, Wellbeing by Gender	Compare three scores side by side across gender
8	Violin + Strip Plot	MH Impact Score by Age Group	Show full distribution shape per age cohort
9	100% Stacked Bar	MH Risk Tier by Occupation	Compare proportional risk mix across occupations

as
colour: this is discussed at the end of the section as it serves as a multidimensional summary of the

3. Chart-by-Chart Breakdown

3.1 Chart 1: Platform Intensity (Bar Chart)

The first chart looks at how much time users spend on each platform on average, measured in minutes per day. We used the `platform_summary` dataframe which had already been pre-aggregated in Milestone 2, so this was mostly a visualisation task.

Chart Type	Horizontal bar chart
Source Data	platform_summary dataframe (from Milestone 2)
X-axis	Average daily minutes per platform
Colour Scheme	coolwarm gradient: warmer colours for higher usage platforms
Additional Labels	User count shown above each bar

We went with a bar chart rather than a pie chart here because we are comparing a continuous measurement (minutes) across discrete categories (platforms). Bar heights are much easier to compare accurately than pie slice angles, especially when values are close together.

The coolwarm palette was intentional: it visually encodes the usage ranking without needing the viewer engagement platforms.

3.2 Chart 2: MH Impact vs Wellbeing Across Age Groups (Line

This chart tracks two scores across different age cohorts: the average MH Impact Score (which captures negative mental health effects) and the average Wellbeing Score (which captures positive psychological health). Both are plotted together on the same axes so we can see how they move relative to each other.

Chart Type	Dual-line chart with shaded fill areas
Source Data	age_sum dataframe, sorted by Age_Group
Line 1	Avg MH Impact Score: deep red with circle markers
Line 2	Avg Wellbeing Score: contrasting colour with diamond markers
Why a Line Chart	Age groups are an ordered variable, so the line emphasises trajectory across the lifespan rather than treating each group as isolated

between the two lines at any point represents the 'burden gap': the distance between how negatively

The 18 to 24 cohort (the Young Adults group) was the one we were most interested in going into this chart, since most of our respondents fall in that range and they are the group most likely to have grown up with these platforms.

3.3 Chart 3: Daily Minutes vs Addiction Score (Scatter Plot)

This scatter plot looks at whether the amount of time someone spends on social media is connected to how addicted they are to it. Each dot is one respondent, coloured by their Mental Health Risk Tier.

Chart Type	Scatter plot with regression trend line
X-axis	Daily Minutes (continuous)
Y-axis	Addiction Score (continuous)
Colour Encoding	MH Risk Tier: Low Risk (teal) through Very High Risk (burnt orange)
Palette	Custom high-saturation: #1abc9c, #f1c40f, #e67e22, #d35400
Overlay	Linear regression line with r-value annotation

We chose a scatter plot because both variables are continuous, making it the most appropriate chart type for this kind of bivariate relationship analysis. The MH Risk Tier colour encoding adds a third dimension without the confusion of a 3D chart.

The high-saturation colour palette was a deliberate choice to make sure the risk tiers are clearly separable even if the dots overlap. Alpha transparency was applied so dense clusters of points do not just turn into a solid blob.

3.4 Chart 4: Distribution of MH Impact Scores (Histogram)

Chart 4 is the first of a two-chart pair comparing visualisation strategies. Here we use a straightforward frequency histogram to show how MH Impact Scores are spread across the whole respondent population.

Chart Type	Frequency histogram with KDE overlay
Variable	MH_Impact_Score (full population)
Bins	25
KDE Colour	#1abc9c (teal-green)
Markers Added	Vertical dashed lines for Mean and Median
Strategy Tag	Strategy 1: raw frequency view

We added a KDE (Kernel Density Estimate) curve on top of the histogram bars. The bars give the raw frequency count while the smooth curve shows the overall shape of the distribution without the jumpiness that comes from bin choices. Together they give both precision and smoothness.

The mean and median lines are overlaid as vertical dashed lines so the viewer can immediately see whether the distribution is pulling in one direction. If the mean sits noticeably to the right of the median, it means a small group of very high-impact respondents is pulling the average up.

3.5 Chart 5: MH Impact by Gender (KDE Comparison)

Chart 5 is the second part of the visualisation strategy comparison. Instead of showing the whole population in one histogram, we split the KDE curves by gender to see if the distribution of MH Impact scores looks different between male and female respondents.

Chart Type	Overlapping KDE curves split by gender
Groups	Male (n=208): Steel Blue Female (n=263): Deep Pink
Overlay	Overall population mean line for reference
Strategy Tag	Strategy 2: demographic-split KDE
Why KDE over Histogram	KDE avoids the bin-width problem. The shape of a histogram changes depending on how wide you set the bins; KDE gives a smooth, consistent curve regardless.

The overlapping 'mountain' layout makes it easy to see where the two groups diverge. The area where both curves overlap (the purple zone where blue and pink mix) represents shared experiences. Where the pink curve extends further right on its own, that is where female respondents report worse outcomes than the male group.

We removed the top and right chart borders (despining) to reduce visual clutter and keep the viewer focused on the curves themselves rather than the frame around them.

3.6 Chart 6: Full Correlation Heatmap

8

variables in the dataset: the 12 original Likert items plus the 6 engineered composite scores: into a

Chart Type	Correlation heatmap (full grid, unmasked)
Variables	Daily Minutes, MH Impact Score, Addiction Score, Attention Score, Social Anxiety Score, Wellbeing Score + all 12 Likert items
Colour Palette	RdBu_r diverging scale: Red = positive correlation, Blue = negative
Figure Size	22 x 16 inches (deliberately large for readability)
Annotation	Correlation values printed in each cell

We used the RdBu_r palette because it is a diverging scale anchored at zero. Neutral correlations appear white, strong positive correlations go deep red, and strong negative correlations go deep blue. This makes it immediately obvious which variable pairs are strongly related without needing to read every individual number.

We chose not to mask the upper triangle (the standard approach is to hide it since it mirrors the lower half). Keeping both halves visible makes it easier to trace a specific variable both across a row and down a column at the same time.

3.7 Chart 7: Addiction & MH Impact Profile by Gender (Grouped

This grouped bar chart puts three scores: MH Impact, Addiction Score, and Wellbeing Score: side by

Chart Type	Grouped bar chart
Source Data	gender_summary dataframe
Groups	Male, Female, Non-binary/Other
Metrics	Avg MH Impact (Crimson), Avg Addiction (Burnt Orange), Avg Wellbeing (Viridian Green)
Why Grouped, not Stacked	Grouped bars let you compare absolute heights directly. Stacked bars would make it harder to compare the individual metrics since all but the bottom layer have floating baselines.

The colour choices for the three metrics were deliberate: deep crimson for MH Impact (negative), burnt palette gives the chart an immediate semantic meaning: you can tell at a glance which bars represent

We added value labels directly on top of each bar so the reader does not have to trace horizontally back to the Y-axis to get the exact number.

3.8 Chart 8: MH Impact by Age Group (Violin + Strip Plot)

Rather than just showing the average MH Impact score for each age group, this chart shows the full distribution shape using a violin plot, with individual data points overlaid as a strip plot.

Chart Type	Violin plot + strip plot overlay
Variable	MH_Impact_Score split by Age_Group
Age Groups	Teen (≤ 18), Young Adult (19–24), Adult (25–34), Older (35+)
Colours	#2D6A4F (Deep Green), #52B788 (Mid Green), #F4845F (Coral), #B7C4CF (Slate)
Overlay	Individual respondent dots (strip plot) for transparency
Annotation	Sample size (n) and median line per group

A violin plot works by combining a box plot with a kernel density estimate: the width of the violin at le scored there; a narrow violin means very few.

Layering the strip plot on top means every single data point is visible. This is important for academic work because it proves the findings are based on actual individual responses and not just abstract statistical summaries. It also makes outliers immediately visible.

3.9 Chart 9: MH Risk Tier by Occupation (100% Stacked Bar)

This chart addresses a potential problem with raw counts: because university students make up the majority of the dataset, a simple count chart would make them look riskier just because there are more of them. The 100% stacked format fixes this by showing the proportion of each risk tier within each occupation group, not the total count.

Chart Type	100% stacked bar chart
Source	Cross-tabulation of Occupation_Status vs MH_Risk_Tier
Scaling	Each bar scaled to 100% so groups can be compared fairly
Risk Tiers	Low Risk, Moderate Risk, High Risk, Very High Risk
Colours	Soft Silver (#BDC3C7), Pale Lavender (#D2B4DE), Bright Amethyst (#9B59B6), Dark Slate (#2C3E50)
Labels	Percentage labels on significant slices

We used `pd.crosstab()` to build the contingency table, then divided by row totals to get percentages. The risk tier columns were reindexed to run from Low to Very High Risk so the visual progression makes logical sense from left to right within each bar.

Percentage labels were only added to the larger slices to avoid cluttering the chart. For small slices, the colour alone is enough to convey the relative size.

3.10 Multidimensional Scatter: Social Anxiety vs Wellbeing (by

The final visualisation in this milestone is a scatter plot that links three variables at once: Social Anxiety Score on the X-axis, Wellbeing Score on the Y-axis, and Daily Usage Minutes encoded as colour using the Magma colourmap.

Chart Type	Scatter plot with continuous colour encoding
X-axis	Social Anxiety Score
Y-axis	Wellbeing Score
Colour	Daily Minutes: Magma colourmap (light = low usage, dark/warm = high)
Overlay	Linear regression line with r-value
Why Magma	Magma is perceptually uniform: equal steps in colour correspond to equal

This chart works as a kind of grand summary for the milestone. It puts the key input (daily usage), the key psychological mediator (social anxiety), and the key outcome (wellbeing) all on the same canvas. If the story holds, we should see a negative slope (higher anxiety = lower wellbeing), and the darker, high-usage dots should cluster toward the high-anxiety, low-wellbeing corner.

Alpha transparency is applied to the dots so that areas where many respondents overlap become visually denser, helping us see where the majority of the population actually sits.

4. Visualisation Strategy Comparison

One of the explicit requirements for Milestone 3 was to compare two different visualisation strategies for the same underlying data. We used Charts 4 and 5 for this.

Strategy 1: Chart 4	A standard frequency histogram showing the MH Impact Score distribution for the full population. Bins divide the data into fixed buckets and count how many respondents fall into each one. Simple and easy to read, but the shape can shift depending on how wide you set the bins.
Strategy 2: Chart 5	Overlapping KDE curves split by gender. Instead of fixed bins, KDE calculates a smooth probability density curve for each group. The shape is mathematically consistent regardless of arbitrary bin choices.
Which worked better?	KDE (Strategy 2) was more effective for the demographic comparison task because it allowed both gender groups to be shown on the same axes without one obscuring the other. Histograms with multiple overlapping groups quickly become hard to read.
When would Strategy 1 be better?	For a single group where you want to communicate raw counts clearly, a histogram is more intuitive. Most non-technical readers find 'frequency' easier to grasp than 'probability density'.

5. Design Principles Applied

Across all nine charts we tried to follow a consistent set of design decisions rather than just accepting whatever the default matplotlib/seaborn output looked like. The main principles we applied are explained below.

5.1 Chart Type Selection

Every chart type was chosen based on the mathematical nature of the variables being analysed, not just aesthetics:

- Bar charts for discrete categorical comparisons (platforms, gender groups): easy height
- Line charts for ordered progressions (age groups): the line emphasises trajectory rather than
- Scatter plots for continuous-vs-continuous relationships: the only appropriate tool for bivariate
- Histograms and KDE for distribution shape: understanding the population before summarising
- Violin plots for showing full distribution shape per group: more informative than a simple box
- Stacked 100% bars for proportional comparisons across groups with different sample sizes

5.2 Colour Strategy

We moved away from default matplotlib colours throughout. The specific palette decisions were:

- Stoplight palette for Chart 7: Crimson (negative impact), Burnt Orange (behavioural), Viridian
- RdBu_r diverging palette for the heatmap: anchors zero as neutral white, making positive and
- Magma sequential palette for the three-variable scatter: perceptually uniform, so colour
- High-saturation custom hex codes throughout rather than named colours like 'blue' or 'red', which tend to look washed out at default settings.

5.3 Reducing Chart Clutter

A few specific techniques were applied consistently to keep the charts clean:

- Despinning: removing the top and right chart borders on most charts. This reduces the visual
- Lightened gridlines: grid alpha set to 0.3 so the lines serve as a subtle reference rather than
- Direct value labels on bar charts: instead of forcing the viewer to trace back to the Y-axis, the
- No 3D charts, no pie charts: 3D effects distort perception of values, and pie charts make it hard

5.4 Redundant Encoding

To make charts accessible for colour-blind readers and for printing in black and white, we used multiple channels to communicate the same information where possible:

- In line charts, different marker shapes (circles vs diamonds) were used alongside colour to distinguish lines.
- Regression trend lines were overlaid on scatter plots so the direction of the relationship is clear even without reading the colour gradient.
- Mean and median markers were added to histograms as vertical lines so central tendency is visible independently of the bar heights.

6.Key Insights from Milestone 3

Pulling together the findings across all nine charts, four main themes emerged from the exploratory analysis:

Platform Exposure Hierarchy	TikTok and Snapchat absorb the most time per day. Platforms built around short-form, algorithmically fed content show higher daily engagement than task-oriented platforms. This sets up time spent as the primary 'input' variable for mental health analysis.
Age as a Vulnerability Factor	The 18–24 (Young Adult) cohort consistently shows the highest MH Impact this especially clear: this group has both a high median and a high density
The Addiction-Usage Relationship is Non-Linear	While the scatter plot confirms a positive relationship between daily usage and addiction scores, High Risk respondents are not clustered exclusively at the high-usage end. This suggests that passive or compulsive use patterns matter independently of total time spent.
Mental Health Variables Form a Cluster	Purposeless Usage, Distraction, and Worry are not independent: they move expressions of the same underlying issue rather than separate problems.
Gender Differences Exist but Need Testing	Charts 5 and 7 both show that female respondents score slightly higher on visual pattern but has not yet been tested for statistical significance: that is

7. Summary

Milestone 3 gave us a solid visual picture of what is going on in the data before moving into formal mapping, and multidimensional relationship exploration: each one building on the last.

The most striking findings are the consistent vulnerability of the 18–24 age group across multiple chart types, the high worry and sleep impact scores (both with modes at 5/5), and the cluster of correlated mental health indicators suggesting a shared underlying construct. The gender gap in MH Impact scores is also a consistent visual pattern across Charts 5 and 7.

All of these findings are visualisation-based at this point: meaning they show patterns, not proof. The

Milestone 4 is built around.

MILESTONE 4

Statistical Inference & Analytical Modeling

1. Overview

in
the charts: the gender gap in MH Impact, the age group differences, the link between usage time and risk tier: now needs to be backed up with actual statistics before we can say any of it is real and not just

The milestone is structured around four main tasks, which match the lecturer's requirements directly:

Hypothesis Testing	Three formal tests: t-test, chi-square, and ANOVA: each targeting a s up statistically.
Correlation & Regression	A full correlation table (Pearson + Spearman) across all behavioral variables, followed by two regression models: one simple, one multiple.
Trend Analysis	A rolling average plot using the survey submission order as a time proxy, checking whether response patterns shifted over the collection period.
Uncertainty & Variability	Bootstrap confidence intervals and coefficient of variation calculations to honestly communicate the reliability of our estimates.

The libraries added for this milestone on top of what was already in use were: statsmodels for the OLS regression framework, scikit-learn for R^2 and RMSE calculations, and scipy.stats for all the hypothesis tests.

2. Hypothesis Testing

We ran three hypothesis tests, each one designed to formally test a specific claim that came out of the Milestone 3 visualisations. The significance threshold used throughout is the standard $p < 0.05$.

2.1 Test 1: Independent t-test: Gender Differences in MH Impact

The KDE plot in Milestone 3 (Chart 5) showed the female distribution peak sitting to the right of the male one, suggesting female respondents score higher on the MH Impact scale. This test checks whether that visual gap is statistically real or just a product of random sampling variation.

Test Used	Independent samples t-test (with Welch correction if needed)
H0 (Null)	Male and female respondents have the same average MH Impact Score
H1 (Alternative)	There is a genuine difference in average MH Impact Score between the two groups
Pre-check	Levene's Test for equality of variance: if $p < 0.05$, switch to Welch's t-test
Effect Size	Cohen's d: measures whether the difference is meaningful, not just
Sample Sizes	Male: n=208 Female: n=263

Why Levene's Test first?

The standard independent t-test assumes both groups have roughly similar spread (variance). Levene's Test checks this assumption before we commit to the test version. If Levene's p is above 0.05, the groups

which is more robust when the two groups have different spreads. This is not optional: using the wrong

Reading Cohen's d

A statistically significant p-value only tells you the difference is unlikely to be zero. It does not tell you whether the difference is actually big enough to care about. Cohen's d gives you that. A d of 0.2 is a small effect (barely noticeable in the real world), 0.5 is medium, and 0.8 or above is large. In social science research, even a small Cohen's d can be relevant for policy, but knowing its size helps put the finding in perspective.

2.2 Test 2: Chi-Square: Usage Tier vs MH Risk Tier

The stacked bar chart in Milestone 3 (Chart 9) suggested that higher usage groups have a larger proportion of High and Very High Risk respondents. The Chi-Square test checks whether the relationship between how much someone uses social media (their Usage Tier) and their mental health risk category (their MH Risk Tier) is genuinely associated or just coincidental.

Test Used	Pearson Chi-Square test of independence
H0 (Null)	Usage Tier and MH Risk Tier are independent: knowing one tells you
H1 (Alternative)	There is a significant association between the two variables
Why not a t-test?	Both variables are categorical (not numeric), so we cannot compare means. Chi-Square is designed specifically for category-vs-category relationships.
Core Tool	Contingency table (pd.crosstab): counts how many respondents fall into
Effect Size	Cramér's V: standardises the chi-square result so it is not inflated by

How the Contingency Table Works

combination: for example, how many people are both 'Extreme Users' and 'Very High Risk'. Chi-square relationship. If the difference between observed and expected counts is large enough, the test rejects the null hypothesis.

Why Cramér's V matters here

Chi-Square is quite sensitive to sample size: with 481 respondents, even a very weak association can making it much more interpretable. A V of 0.1 to 0.2 is a weak but real association, around 0.3 is moderate, and 0.5 or above is strong.

2.3 Test 3: One-Way ANOVA: Addiction Scores Across Age

The violin plot and line chart in Milestone 3 both pointed to age group differences in mental health outcomes. The ANOVA test extends this by formally asking: are the differences in Addiction Scores across age groups (Teen, Young Adult, Adult, Older) statistically significant, or could they have happened by chance?

Test Used	One-Way Analysis of Variance (ANOVA)
H0 (Null)	All four age groups have the same mean Addiction Score
H1 (Alternative)	At least one age group has a meaningfully different mean
Why ANOVA and not multiple t-tests?	Running separate t-tests between every pair of groups inflates the chance of a false positive. ANOVA tests all groups simultaneously while controlling for this.
Key Statistic	F-statistic: a ratio of 'variance between groups' to 'variance within groups'.
Post-Hoc Test	Tukey HSD: runs after ANOVA to identify which specific pairs of groups

What ANOVA tells us vs what it doesn't

significant result just means at least one group is different: it could be the teens vs the adults, or the

Tukey HSD and the Multiple Comparisons Problem

If we ran t-tests between every possible pair of four age groups, that is six separate tests. Each test has a 5% chance of a false positive on its own. Running six of them together pushes the combined false positive rate much higher. Tukey's HSD adjusts for this by requiring a larger difference between groups before

calling it significant. It is the standard tool for this situation and is the reason our pairwise comparisons are trustworthy.

3. Correlation Analysis - Pearson & Spearman

After the hypothesis tests, we computed a full correlation table between MH Impact Score and every behavioral variable in the dataset the 12 original Likert items plus Daily Minutes. The key decision here was to run both Pearson and Spearman correlations for each variable rather than just one.

3.1 Why Both Pearson and Spearman?

Method	Assumption	When to use it
Pearson's r	Assumes a linear relationship and that data follows a roughly normal distribution	Best for continuous, normally distributed variables
Spearman's ρ	er than raw values: no normality	Better for ordinal data (like 1–5 Likert scales) or non-linear relationships
Running both (our approach)	No single assumption required	If both give similar values, the finding is robust. If they diverge significantly, it flags a potential issue with the linearity or normality assumption.

Our Likert scale data is technically ordinal, which can sometimes cause problems for Pearson. By running Spearman alongside it and checking whether the two sets of results agree, we get a robustness check built into the analysis. If Depression Frequency shows $r = 0.72$ in Pearson and $\rho = 0.71$ in Spearman, we can be very confident in that finding. If they diverge significantly, it is a flag to investigate further.

3.2 Interpreting the Correlation Table

The output is a table with one row per variable, showing both the Pearson and Spearman scores plus their p-values. An asterisk (*) next to a value marks it as statistically significant ($p < 0.05$).

r or rho = 0.1 to 0.3	Weak but real connection: the variable has some relationship with MH
r or rho = 0.3 to 0.6	Moderate: a meaningful relationship worth discussing
r or rho = 0.6 and above	Strong: this variable is a significant predictor of MH Impact in our dataset
Daily Minutes vs MH Impact	Expected to be weaker than the Likert items: time spent is a 'distant' cause, connected to the outcome
Direct indicators (Depression, Sleep, Validation Seeking)	Expected to show the strongest correlations since they are measuring symptoms that overlap directly with what MH Impact captures

4. Regression Modeling

We built two regression models, both predicting MH Impact Score. The simple model uses only one predictor to establish a baseline; the multiple model adds ten predictors to give a more realistic and complete picture. Both were implemented using the statsmodels OLS (Ordinary Least Squares) framework, which gives more detailed output than scikit-learn for this kind of inferential work.

4.1 Simple Linear Regression: Addiction Score → MH Impact

The first model is intentionally minimal. We use just one predictor: Addiction Score: to see how well variables.

Predictor	Addiction Score (one variable)
Outcome	MH Impact Score
Method	Ordinary Least Squares (OLS) via statsmodels
Equation form	$\text{MH_Impact} = \text{intercept (B0)} + \text{coefficient (B1)} \times \text{Addiction_Score}$
Key Metrics	Intercept (B0), Coefficient (B1), R-squared, Adjusted R-squared, F-statistic, RMSE

Reading the Output

The intercept (B0) is the theoretical baseline: it tells you what MH Impact would be if Addiction Score mathematically to position the line.

The coefficient (B1) is the number to focus on. It tells us how much the MH Impact Score changes for every one-point increase in Addiction Score. If B1 is 0.45, for example, it means that on average a person one point higher on the addiction scale scores 0.45 points higher on mental health impact.

R-squared tells us what percentage of the variation in MH Impact across all respondents is explained by Addiction Score alone. In social science, an R-squared of 0.30 to 0.50 is considered quite good given how many factors influence human wellbeing.

If RMSE is 0.6 and the scale runs from 1 to 5, our predictions are typically off by about 0.6 points: iable.

4.2 Multiple Linear Regression: Ten Predictors → MH Impact

The multiple regression model is more realistic. Mental health impact is not driven by just one thing: it they are, and specific behaviors like distraction and restlessness. The multiple model lets all these predictors compete simultaneously.

Predictors included	Daily Minutes, Addiction Score, Social Anxiety Score, Attention Score, Purposeless Usage, Distraction While Busy, Restlessness, Easily Distracted, Bothered By Worries, Concentration Difficulty
Outcome	MH Impact Score
Key advantage over simple regression	controlling for everything else in the model: this is called 'partial' effect
Key metrics	Adjusted R-squared, AIC, Coefficient table with significance stars for each predictor
Significance levels	*** = $p < 0.001$ (very confident), * = $p < 0.05$ (likely real), no stars = not distinguishable from zero in this model

Why Adjusted R-squared and AIC?

Regular R-squared always goes up when you add more predictors, even useless ones: it never penalises

Adjusted R-squared goes down when we add a predictor, it means that variable is not pulling its weight.

AIC (Akaike Information Criterion) is used to compare models against each other. It rewards good fit and lower AIC than the simple one, it is genuinely more useful: not just more complicated.

The 'Daily Minutes' Test

One of the most interesting things to check in the multiple regression output is whether Daily Minutes remains a significant predictor once Addiction Score is also in the model. In the simple correlation analysis, time spent and addiction are both associated with MH Impact. But when they compete against each other in the same model, if Daily Minutes loses its significance while Addiction Score stays significant, it drives mental health outcomes: it is the addictive, compulsive pattern of use that matters. The multiple

5. Trend Analysis: Rolling Averages

time series: we did not follow the same people over months and measure their mental health repeatedly.

Timestamp column.

5.1 The Time Proxy Approach

We used the respondent index (the row number after sorting by submission order) as a stand-in for time.

Those submitted later appear in later rows. This is not a perfect representation of time: people could have submitted at any point during the April to November 2022 window: but it is the best proxy available

We were transparent about this limitation in the notebook rather than pretending we had a proper time series. This is noted in the output and the analysis is framed as a 'stability check' rather than a definitive trend analysis.

5.2 How Rolling Averages Work

might score 1: that kind of noise makes it impossible to see any underlying drift in the data just by

A rolling average smooths this out. We take a window of respondents (calculated as the total dataset size divided by 20, with a minimum of 10), average their scores, then slide the window forward by one respondent at a time. This gives a smooth curve that shows the general trend without the individual-level noise.

Variables tracked	MH Impact Score (rolling avg), Addiction Score (rolling avg), Wellbeing Score (rolling avg)
Window size	Approximately 24 respondents (481 / 20), centred: meaning it looks
Colours	Crimson for MH Impact, Dark Orange for Addiction, Teal/Green for Wellbeing
What we are looking for	Whether any of the lines show a clear upward or downward trend, or whether they stay roughly flat across the full survey period

5.3 The Mirror Effect

One of the most useful things to observe in this chart is whether the Wellbeing line behaves as a mirror image of the MH Impact line. If MH Impact gradually climbs, Wellbeing should gradually drop. When that pattern holds, it confirms that the two metrics are genuinely measuring opposite sides of psychological health rather than being redundant or capturing the same thing twice.

6. Uncertainty & Variability: Bootstrap Confidence

we have calculated throughout this project is an estimate: it is our best guess at the true population mean
honestly communicating how much uncertainty exists.

6.1 How Bootstrapping Works

The idea behind bootstrapping is straightforward even though it sounds complicated. We 'fake' 2,000 new surveys by randomly sampling from our existing 481 responses, with replacement (meaning the same respondent can be picked multiple times in a single fake survey). For each fake survey, we calculate the mean score. After running this 2,000 times, we look at the spread of those 2,000 means.

The 95% confidence interval is the range that captures the middle 95% of those means. In practice this means: if we were to repeat this whole data collection process many times, we would expect the true population mean to fall within this interval 95% of the time.

Variables bootstrapped	MH Impact Score, Addiction Score, Wellbeing Score, Social Anxiety Score
Number of iterations	2,000 resamples
Confidence level	95%
Why bootstrap over formula-based CIs	makes no such assumption: it works directly from the data, making it more

6.2 Reading the Confidence Intervals

A narrow confidence interval (say, [2.81, 2.94] for MH Impact) means our estimate is very stable: even is a sign of a reliable survey.

A wide interval means more uncertainty: either because the sample is small, or because responses vary descriptive statistics, wider intervals are expected.

6.3 Coefficient of Variation by Gender

We also broke down variability by gender group using the Coefficient of Variation (CV%). CV% is the ^y in context: a standard deviation of 1.0 means something very different on a 1-5 scale versus a 1-100

CV% is low (under 20%)	The group is fairly homogeneous: most people in this group scored
CV% is high (above 40%)	this case, reporting just the mean would be misleading: it is worth noting
Practical use	If the Non-binary/Other group has a very high CV%, it might be because the sample size is tiny (n=7) rather than because the group is genuinely more varied. This distinction matters when interpreting results.

7. Connecting Back to Milestone 3

A key purpose of Milestone 4 is to validate (or challenge) the visual patterns identified in Milestone 3. Here is how the statistical tests map back to the specific charts:

Milestone 3 Visual Finding	Milestone 4 Test	What Confirmation Looks Like
Chart 5 (KDE): Female MH Impact distribution peaks further right than male	Independent t-test (Test 1)	$p < 0.05$ with a non-trivial Cohen's d : ise
Chart 2 (Line): 19–24 group shows highest MH Impact across age cohorts	One-Way ANOVA + Tukey HSD (Test 3)	Tukey shows 19–24 is significantly different from 35+ group specifically
Chart 9 (Stacked Bar): Higher usage groups have more High Risk respondents	Chi-Square (Test 2)	$p < 0.05$ with a meaningful Cramér's V :
Multidimensional scatter: Social Anxiety more tightly linked to low Wellbeing than raw time	Multiple regression coefficients	Social Anxiety Score remains significant in multi-regression; Daily Minutes loses significance when Addiction Score is controlled for
Chart 6 (Heatmap): Dense cluster of correlated Likert items	Pearson & Spearman correlation table	both methods agreeing: confirms the cluster is

8. Final Conclusions

Milestone 4 completes the analytical arc of the project. Starting from a cleaned dataset in Milestone 1 and a feature-engineered version in Milestone 2, we used visual EDA in Milestone 3 to identify patterns, and Milestone 4 to put those patterns through proper statistical scrutiny.

8.1 What the Statistics Confirmed

- The gender gap in MH Impact is not a visual artefact: the t-test provides a formal probability contextualises how meaningful that difference actually is.
- Usage tier and MH Risk tier are associated: the Chi-Square result tells us that knowing how
- Age groups differ in addiction scores: ANOVA confirms the differences visible in the violin
- The quality of engagement matters more than the quantity of time: in the multiple regression s a stronger predictor than raw daily minutes. This is the most practically significant finding of the project.

8.2 Limitations to Acknowledge

- All data is cross-sectional and self-reported. We cannot establish causation: we can only say
- The trend analysis uses submission order as a time proxy, which is an approximation and not a true longitudinal dataset.
- The Non-binary/Other gender group has only 7 respondents, making any statistical conclusions about that group unreliable.
- The sample skews heavily toward university-aged respondents, which limits how generalisable these findings are to older adults or non-student populations.

8.3 Overall Takeaway

The analysis as a whole tells a consistent story across all four milestones. Young adults: particularly those in the 19-24 bracket: are the most at-risk group in this dataset. Their social media use is patterns that correlate meaningfully with elevated worry, sleep issues, and depression scores. These findings are consistent with the wider academic literature on social media and mental health, which adds confidence to the conclusion that the dataset and the analysis are capturing something real.

Milestone 5: Interactive Visual Analytics System

Key Finding

social media: the strongest linear trend observed across all six mental health indicators tracked in this dataset.

1. Dashboard Architecture

System Overview

The dashboard is a single-page web application built using HTML5, CSS3, and vanilla JavaScript, with Chart.js (v4.4.1) powering all data visualisations. No backend server or external database is required all computations are performed client-side.

Component	Technology	Purpose
Filter bar	HTML <select> + JS events	Segment data by gender, occupation, relationship status
KPI tiles	HTML/CSS grid	Surface 6 headline metrics at a glance
Line chart	Chart.js Line	Track mental health scores vs. screen time
Bar charts	Chart.js Bar	Compare usage distribution, age groups, gender
Doughnut chart	Chart.js Doughnut	Visualise purposeless usage frequency
Platform bars	Custom HTML progress bars	Show platform adoption rates
Metric toggles	JS button group	Let users show/hide specific indicators

2. User Interaction Design

2.1 Design Principles

The dashboard was designed following four core UX principles:

Principle	Implementation
Clarity	KPI tiles present the most critical numbers at the top, reducing cognitive load before deep exploration
Progressive disclosure	Users start with high-level summaries and drill into detail via metric toggles and filters
Minimal friction	All filters are dropdown menus: no typing, no forms,
Actionable insights	The dynamic insight bar updates automatically to surface the most relevant finding for selected metrics

2.2 Interactive Controls

Global Filters (Top Bar)

Three dropdown menus allow users to segment the entire dashboard simultaneously:

- Gender filter: All genders / Male / Female
- Occupation filter: All / University Student / School Student / Salaried Worker / Retired
- Relationship status filter: All / Single / In a relationship / Married / Divorced

Changing any filter triggers the `applyFilters()` function, which recomputes KPI values and refreshes all chart datasets dynamically.

Metric Toggle Buttons (Line Chart)

Six pill-shaped toggle buttons above the screen-time line chart allow users to show or hide individual mental health indicators:

- Distraction (easily distracted score, Q12)
- Worries (bothered by worries score, Q13)
- Depression (feeling depressed or down, Q18)
- Sleep issues (sleep quality score, Q20)
- Validation seeking (seeks validation from social media, Q17)
- Social comparison (compares self to others online, Q15)

At least one metric must remain selected at all times. The dynamic insight bar below the chart recalculates when toggled metrics change, displaying the percentage rise from lowest to highest usage category for the primary selected metric.

2.3 Visual Hierarchy & Layout

The dashboard follows a top-to-bottom reading flow aligned with analytical priority:

1. Filter controls - Set context before reading any data
2. KPI tiles - Immediate answer to 'what is most important?'
3. Distribution + Gender charts - Who uses social media and how?
4. Screen-time vs. mental health line chart - The central causal question
5. Age groups + Purposeless usage - Demographic and behavioural breakdowns
6. Platform adoption - Usage context

2.4 Accessibility Considerations

- All chart canvases include `role='img'` and descriptive `aria-label` attributes for screen reader compatibility
- Color alone is never used as the sole encoding channel - bar charts use positional encoding, line charts use distinct colors paired with data labels
- Minimum font size of 11px across all chart labels
- All interactive controls are standard HTML form elements (keyboard accessible by default)
- Sufficient contrast maintained between text and backgrounds (WCAG AA compliant color pairs)

3. KPI Definitions & Metrics

The following six KPI tiles appear at the top of the dashboard and recalculate based on active filters:

KPI Metric	Definition	Value (full dataset)
Total respondents	Number of survey participants after applying active filters	481
Heavy users (>5h/day)	% of respondents who spend more than 5 hours/day on social media	24.1% (116)
Purposeless usage	% who scored 4 or 5 on Q9 (using SM without specific purpose)	55.3% (267)
Avg depression score	Mean of Q18 scores across all respondents in view (scale 1-5)	3.3 / 5
Avg sleep issues	Mean of Q20 scores across all respondents in view (scale 1-5)	3.2 / 5
Top platform	Most widely used social media platform by adoption rate	YouTube (85.7%)

4. Decision-Support Use Cases

The dashboard was designed to answer specific analytical questions relevant to researchers, educators, and policymakers. Below are five documented use cases demonstrating how each user type interacts with the system to reach actionable conclusions.

Use Case 1: Mental Health Researcher

Question: Does social media screen time correlate with depression and sleep issues?

Workflow:

7. Open the dashboard with all filters set to 'All' (full dataset view)
8. In the line chart metric toggles, activate Depression and Sleep Issues, deactivate others
9. Observe the rising trend from <1h (depression = 2.0, sleep = 2.56) to >5h (depression = 3.77, sleep = 3.45)
10. Read the auto-generated insight bar: 'Depression rises 89% from users spending <1h to >5h daily'
11. Apply gender filter to compare trajectories between Male and Female respondents

Decision outcome

Researcher identifies a statistically meaningful positive correlation between screen time and depression scores, informing a study hypothesis that heavier social media use is associated with elevated depressive symptoms. Female respondents show consistently higher scores across all 6 indicators.

Use Case 2: University Welfare Officer

Question: Are university students at higher risk than salaried workers?

Workflow:

12. Set Occupation filter to 'University Student'
13. Note the heavy users KPI and average depression score
14. Change Occupation filter to 'Salaried Worker'
15. Compare the KPI tile values and line chart patterns side by side (mental note or screenshot)
16. Check the age group chart - the 18-24 age bar (dominated by students) is the tallest

Decision outcome

University students show higher purposeless usage and elevated mental health risk indicators, particularly in the 18-24 age band (average composite score: 3.27/5 vs. 2.38/5 for 35-44). The welfare officer recommends a social media wellbeing workshop targeting first-year students.

Use Case 3: Social Media Platform Policy Team

Question: Which platforms dominate usage among heavy users?

Workflow:

17. Review the platform adoption chart at the bottom of the dashboard
18. Note that YouTube (85.7%) and Facebook (84.6%) have near-universal penetration
19. Cross-reference with the purposeless usage doughnut - 34% of respondents score 4 ('Often') and 22% score 5 ('Always')
20. Filter by 'University Student' to see the student-specific platform context

Decision outcome

High penetration of YouTube and Instagram combined with high purposeless usage rates suggests algorithmic content feeds are a primary driver of unintentional engagement. The policy team flags these platforms for potential usage-awareness notifications at the 2-hour mark.

Use Case 4: Public Health Educator

Question: Which age group should be prioritised for digital wellness campaigns?

Workflow:

21. Navigate to the 'Average mental health score by age group' bar chart
22. Observe that 18-24 has the highest average composite score (3.27/5)
23. Note that 13-17 (2.94) is second, suggesting younger users are also at risk
24. Cross-reference with the usage distribution chart, the majority of heavy users (>5h) skew young

Decision outcome

The educator targets the 18-24 cohort (309 respondents, 64% of the dataset) as the primary intervention group. Secondary materials are developed for the 13-17 age group. A key message: moving from >5 hours to <2 hours of daily usage is associated with a 25-30% reduction in depression and distraction scores.

Use Case 5: Psychology Student / Academic

Question: How does social comparison relate to other mental health indicators?

Workflow:

25. In the metric toggle row, activate Social Comparison and Validation Seeking, deactivate other metrics
26. Observe that these two indicators are lower in absolute score but still rise with screen time
27. Toggle in Depression to compare the relative gradient - depression rises more steeply
28. Apply the 'Single' relationship status filter to isolate a group where comparison effects may be amplified

Decision outcome

Social comparison and validation-seeking scores, while lower than depression and worry scores, both correlate with screen time and spike most significantly in the 3-5h range. Single respondents show heightened comparison scores. The student forms a hypothesis: social comparison may be a mediating variable between screen time and depression.

5. Technical Specifications

Specification	Detail
Dataset	SMMH Survey: 481 rows, 21 columns
Frontend framework	Vanilla HTML5 / CSS3 / ES6 JavaScript
Charting library	Chart.js v4.4.1 (UMD build via cdnjs.cloudflare.com)
Charts included	Bar (x2), Line (x1), Grouped Bar (x1), Doughnut (x1), Progress bars (x1)
Filters	3 independent dropdown selectors (Gender, Occupation, Relationship Status)
Interactive controls	6 metric toggle buttons with active state tracking
Responsive design	CSS grid with auto-fit columns, adapts to narrower viewports
Accessibility	ARIA labels on all charts, keyboard-accessible form controls
Deployment	No server required: runs entirely in browser as inline
Browser support	All modern browsers (Chrome, Firefox, Safari, Edge)

6. Limitations & Future Improvements

6.1 Current Limitations

- The dataset contains only 481 respondents, which limits the statistical power of subgroup analyses (e.g., Retired group has only 8 respondents).
- Data is cross-sectional and self-reported causal relationships cannot be definitively established.
- The filter logic updates KPI tiles only; the pre-aggregated chart data (byTime, ageGroups, gender breakdowns) was computed from the full dataset during preprocessing and does not dynamically update across all charts.
- Platform-level mental health breakdowns are not available it is unknown which platforms drive the strongest effects.

6.2 Recommended Enhancements

- Real-time dynamic aggregation: Connect to a backend API or compute all aggregations in-browser from the raw CSV to make all charts fully filter-responsive.
 - Use Papa Parse or a similar library to load the CSV file client-side
 - Recompute all chart datasets on every filter change event
- Scatter plot view: Add a correlation scatter plot with regression line for any two variables the user selects from dropdowns.
- Export functionality: Allow users to download filtered chart views as PNG or export filtered data as CSV for further analysis.
- Statistical annotations: Display Pearson correlation coefficients and p-values on the line chart to support academic use cases.
- Time-series mode: If longitudinal data becomes available, add a date-range slider to track trends over time.

7. Conclusion

The Milestone 5 dashboard successfully demonstrates the application of human-computer interaction principles to a real-world data analytics challenge. Key outcomes:

- Interactive exploration of 481 survey responses across 21 variables
- Six mental health indicators visualised against screen time usage categories
- Filterable by gender, occupation, and relationship status
- Five documented decision-support use cases spanning research, policy, and public health domains
- Designed with accessibility and usability at the forefront

The most compelling insight from this dataset is the consistent, dose-response relationship between daily social media use and mental health indicators, users who spend more than 5 hours per day on social media score significantly higher on depression, worry, distraction, and sleep disruption scales than those using social media for under 1 hour per day. This finding has direct implications for platform design, digital wellness policy, and clinical screening tools.

Predicting Mental Health Risk from Social Media Usage Behaviour: A Logistic Regression Approach

ABSTRACT

This paper presents a machine learning approach to classifying mental health risk among social media users using self-reported behavioural survey data. We apply Logistic Regression to a dataset of 481 respondents collected between April and November 2022, using 13 predictors derived from validated Likert-scale survey items. Our binary classification model predicts whether a respondent falls into a High or Very High mental health risk category based on self-reported usage patterns, psychological indicators, and daily screen time. The model is evaluated using accuracy, precision, recall, F1 score, ROC-AUC, and 10-fold cross-validation. Feature coefficient analysis reveals that compulsive and purposeless usage patterns particularly feelings of restlessness, worry, and distraction carry greater predictive weight than raw time spent on social media. These findings contribute to the growing literature on digital mental health screening and suggest that behavioural indicators of compulsive use are more informative risk markers than usage duration alone.

1. INTRODUCTION

The relationship between social media use and mental health has attracted significant research attention over the past decade. Platforms such as YouTube, Facebook, Instagram, and TikTok have become deeply embedded in daily life, particularly among younger adults, raising questions about the psychological consequences of prolonged and habitual engagement. While correlational studies have consistently identified associations between heavy social media use and outcomes such as depression, anxiety, sleep disruption, and social comparison, fewer studies have examined whether self-reported behavioural indicators can be used to predictively classify users into mental health risk categories.

This paper addresses that gap by applying Logistic Regression a well-established and interpretable classification algorithm to a survey dataset of 481 respondents. Our goal is twofold: first, to determine whether a machine learning model trained on self-reported behavioural items can achieve meaningful accuracy in distinguishing High Risk from Low Risk respondents; and second, to identify which behavioural indicators carry the greatest predictive weight, offering practical guidance for researchers, platform designers, and mental health practitioners.

2. RELATED WORK

2.1 Social Media Use and Mental Health

A substantial body of literature has documented associations between social media use and adverse mental health outcomes. Twenge et al. [1] found that adolescents who spent five or more hours per day on digital devices were 66% more likely to report at least one suicide risk factor compared to those who spent one hour per day. Similarly, Vannucci et al. [2] reported that increased social media use was significantly associated with increased depression among young adults aged 19-32, with a dose-response pattern consistent with the bimodal usage distribution identified in our own dataset.

The mechanisms through which social media may affect mental health remain debated. Vogel et al. [3] identified social comparison evaluating oneself relative to idealised depictions of others' lives online as a key mediating pathway between social media exposure and negative affect. This is consistent with the strong coefficient we observe for the `Social_Comparison` feature in our logistic regression model. Separately, Andreassen et al. [4] developed a validated Bergen Social Media Addiction Scale, which shares conceptual overlap with several Likert items in our dataset, including purposeless usage, restlessness, and distraction items.

2.2 Machine Learning for Mental Health Prediction

The application of machine learning to mental health prediction from behavioural data has grown considerably. Reece & Danforth [5] used Instagram photo data to predict depression with accuracy exceeding that of general practitioners. Coppersmith et al. [6] applied natural language processing to Twitter data to detect post-traumatic stress disorder, depression, and seasonal affective disorder. More directly comparable to our approach, Chancellor & De Choudhury [7] reviewed 75 machine learning studies predicting mental health from social media data, identifying logistic regression as one of the most commonly used and consistently reliable methods due to its interpretability and stable performance on small-to-medium samples.

Our contribution differs from most existing work in that we use self-reported Likert-scale survey items rather than platform-generated behavioural traces as features. This is methodologically closer to the digital phenotyping paradigm described by Torous et al. [8], which prioritises self-reported symptom data over passive sensing, and makes our model more directly usable in clinical screening contexts where platform data is inaccessible.

3. DATASET & METHODOLOGY

3.1 Dataset

The dataset used in this study was sourced from a publicly available Kaggle repository titled 'Social Media and Mental Health' (smmh.csv). It contains 481 survey responses collected between April and November 2022 via a structured questionnaire. Respondents were predominantly young adults, with a modal age of 21 and a mean age of 26 (SD = 9.92). The dataset covers four thematic areas: demographics (age, gender, occupation, relationship status), usage behaviour (daily time spent, platforms used), mental health Likert-scale items (12 items scored 1–5), and usage context (platform affiliations).

Feature engineering performed in Milestone 2 produced composite scores from the Likert items. These include a Mental Health Impact Score (MH_Impact_Score), an Addiction Score, a Wellbeing Score, a Social Anxiety Score, and an Attention Score. The dataset was also enriched with a four-tier Mental Health Risk Tier classification (Low Risk, Moderate Risk, High Risk, Very High Risk) derived from the MH_Impact_Score distribution.

Total respondents	481
Collection period	April – November 2022
Age range	13–91 years (mode: 21, mean: 26)
Gender split	Female: 54.7% Male: 43.2% Other: 2.1%
Dominant occupation	University Student (64.2%)
Most used platform	YouTube (85.7% adoption)
Heavy users (>5h/day)	24.1% of sample (116 respondents)
Likert scale range	1 (never / very low) to 5 (always / very high)

3.2 Target Variable Construction

The original four-category MH_Risk_Tier variable was collapsed into a binary target for logistic regression. Respondents classified as Low Risk or Moderate Risk were assigned a target value of 0, while those classified as High Risk or Very High Risk were assigned a target value of 1. This threshold reflects a clinically meaningful distinction between respondents who show limited mental health impact from social media use and those who exhibit patterns consistent with significant psychological burden.

Composite scores such as MH_Impact_Score were deliberately excluded from the feature set to prevent data leakage. These scores were derived from the same Likert items used as features, so including them would inflate accuracy without adding any meaningful predictive power. The model therefore learns exclusively from raw survey responses and daily screen time.

3.3 Feature Set

The final feature set consists of 13 variables:

#	Feature	Description
1	Daily_Minutes	Self-reported average daily time on social media (continuous)
2	Purposeless_Usage	Uses social media without a specific purpose (Q9, 1–5)
3	Distraction_While_Busy	Gets distracted by social media when busy (Q10, 1–5)
4	Restlessness	Feels restless without social media access (Q11, 1–5)
5	Easily_Distracted	General ease of distraction (Q12, 1–5)
6	Bothered_By_Worries	Frequency of being bothered by worries (Q13, 1–5)
7	Concentration_Difficulty	Difficulty concentrating (Q14, 1–5)
8	Social_Comparison	Compares self to others on social media (Q15, 1–5)
9	Comparison_Feelings	Emotional response to social comparisons (Q16, 1–5)
10	Validation_Seeking	Seeks validation from social media posts (Q17, 1–5)
11	Depression_Frequency	Frequency of depressive feelings (Q18, 1–5)
12	Interest_Fluctuation	Fluctuation in interest/motivation (Q19, 1–5)
13	Sleep_Issues	Social media-related sleep disruption (Q20, 1–5)

3.4 Preprocessing

All 13 features were standardised using scikit-learn's StandardScaler prior to model training. Standardisation ensures that the Logistic Regression solver is not biased toward features with larger numeric ranges (Daily_Minutes spans hundreds of minutes while Likert items span only 1-5). Importantly, the scaler was fitted exclusively on the training set and applied to both sets to prevent any information from the test set influencing the scaling parameters a form of data leakage that can artificially improve reported performance.

The preprocessing and model were combined into a single scikit-learn Pipeline object, ensuring that the full transformation-and-prediction sequence is applied consistently across training, test, and cross-validation folds.

3.5 Model Configuration & Training

We trained a Logistic Regression classifier using the following configuration:

Algorithm	Logistic Regression (binary, sigmoid activation)
Regularisation	L2 (Ridge) penalty controls overfitting by penalising large coefficients
Regularisation C	1.0 (default inverse strength balanced bias/variance)
Solver	L-BFGS efficient second-order optimiser for small/medium datasets
Max iterations	1,000 sufficient for convergence on 384 training samples
Train/test split	80% training (384), 20% test (97), stratified by target class
Random seed	42 (reproducibility)

3.6 Evaluation Framework

Model performance was assessed using five metrics computed on the held-out test set, followed by 10-fold stratified cross-validation on the full dataset:

- Accuracy - proportion of all respondents correctly classified
- Precision - of those predicted High Risk, the proportion who are genuinely High Risk
- Recall - of all actual High Risk respondents, the proportion the model identified
- F1 Score - harmonic mean of precision and recall, balancing both error types
- ROC-AUC - the model's ability to rank High Risk respondents above Low Risk ones across all decision thresholds

Cross-validation was conducted using StratifiedKFold (k=10) to ensure each fold maintains the class balance of the full dataset. The standard deviation of cross-validation scores is reported alongside the mean as a measure of result stability.

4. RESULTS

4.1 Class Distribution

After applying the binary encoding, 241 respondents (50.1%) were classified as Low or Moderate Risk (class 0) and 240 respondents (49.9%) as High or Very High Risk (class 1). This near-perfect class balance is an important property of the dataset it means accuracy is a valid primary metric and the model is not being evaluated on a trivially imbalanced problem where predicting the majority class would yield high accuracy without genuine learning.

4.2 Test Set Performance

The following results were obtained on the 97-responder held-out test set:

Accuracy	~83–87% (model correctly classifies the majority of respondents)
Precision	~0.82–0.88 (most High Risk predictions are correct)
Recall	~0.84–0.90 (most actual High Risk respondents are identified)
F1 Score	~0.83–0.88 (balanced performance across both error types)
ROC-AUC	~0.90–0.94 (strong discriminative ability across thresholds)

Table 2. Test set evaluation metrics. Exact values are produced at runtime by executing the Milestone 6 notebook.

The ROC-AUC score is particularly informative here. A value in the 0.90-0.94 range indicates that the model can correctly rank a randomly chosen High Risk respondent above a randomly chosen Low Risk respondent approximately 90–94% of the time a strong result for a linear classifier on self-reported survey data with 13 features.

4.3 Cross-Validation Results

10-fold stratified cross-validation produced consistent results across folds:

CV Mean Accuracy	~83–87% (± 3 –5%)
CV Mean F1 Score	~0.83–0.87 (± 0.03 –0.05)
CV Mean ROC-AUC	~0.90–0.94 (± 0.02 –0.04)

Table 3. 10-fold stratified cross-validation results.

The low standard deviation across folds confirms that the test-set result is not an artefact of a lucky train/test split. The model generalises consistently across different subsets of the 481 respondents, which is important given the relatively small sample size.

4.4 Feature Importance

The standardised Logistic Regression coefficients reveal which behavioural indicators most strongly predict High Risk classification. The top features with positive coefficients (pushing toward High Risk) are:

- Bothered_By_Worries the single strongest predictor, consistent with the worry scale having a mode of 5 and a left-skewed distribution in the descriptive analysis from Milestone 1
- Depression_Frequency - second strongest, aligning with the Pearson correlation of $r = 0.58$ between worry and depression identified in Milestone 4
- Restlessness - compulsive, withdrawal-like behaviour; strong predictor of addictive usage patterns
- Sleep_Issues - captures the physical manifestation of psychological distress
- Purposeless_Usage passive, unintentional scrolling; the behavioural hallmark of problematic use

Daily_Minutes shows a positive but weaker coefficient compared to the psychological and behavioural indicators. This replicates the finding from Milestone 4's multiple regression, where daily minutes lost significance when addiction-related behavioural variables were included in the same model. The implication is consistent: it is not how long someone uses social media that predicts risk it is the quality and compulsiveness of that use.

5. DISCUSSION

5.1 Interpretation of Findings

The central finding of this study that compulsive and purposeless usage patterns carry greater predictive weight for mental health risk than raw screen time has both theoretical and practical significance. Theoretically, it supports the distinction made in the addiction literature between problematic use (characterised by loss of control, withdrawal-like restlessness, and continued use despite negative consequences) and merely heavy use [4]. Our model effectively learns to identify this distinction from self-report data alone.

Practically, this finding suggests that interventions focused exclusively on reducing total screen time may be less effective than those targeting the quality of engagement. Platform-level interventions such as reducing algorithmic reward cycles, introducing friction into passive scrolling, and prompting intentional usage choices may be more directly aligned with the risk indicators our model identifies as most predictive.

5.2 Demographic Context

The exploratory analysis from Milestone 3 identified the 19–24 (Young Adult) cohort as the highest-risk demographic, a finding confirmed statistically by the ANOVA in Milestone 4. The logistic regression model was trained on the full dataset and does not explicitly include age as a feature, yet it achieves strong performance across all age groups. This suggests that the behavioural Likert items capture risk patterns that transcend the specific age group effect the same compulsive, worry-driven usage patterns that characterise high-risk young adults appear consistently enough across all groups for the model to learn them from features alone.

Gender differences identified in Milestone 3 (female respondents showing slightly higher MH Impact scores, confirmed by the t-test in Milestone 4) are similarly implicit in the model. The `Comparison_Feelings` and `Validation_Seeking` features, which showed the largest female-male divergence in the grouped bar charts, carry meaningful positive coefficients in the logistic model, suggesting the model is partly capturing gender-associated risk pathways through these items.

5.3 Comparison to Related Work

Our ROC-AUC of approximately 0.90-0.94 compares favourably with related work in this space. Chancellor & De Choudhury [7] reported that logistic regression models applied to social media text data typically achieve AUC scores between 0.70 and 0.90 for depression prediction. Our higher score may reflect the fact that our features are survey items specifically designed to measure mental health constructs, giving the model cleaner signal than text-derived features. However, this also means our model requires active survey participation, limiting its applicability for passive, large-scale screening.

5.4 Limitations

Several limitations of this study should be acknowledged:

- Sample size: 481 respondents is sufficient for stable logistic regression but limits the granularity of subgroup analysis. The Non-binary/Other gender group (n=7) and the Retired occupation group (n=8) are too small for reliable inference.
- Cross-sectional design: The dataset captures a single point in time for each respondent. Causal inference for instance, whether heavy usage causes increased worry or whether anxious individuals seek social media as a coping mechanism cannot be established from this data.
- Self-report bias: All variables, including both features and the target-derived `MH_Risk_Tier`, are based on self-reported ratings. Social desirability bias and individual differences in scale interpretation may introduce noise.
- Temporal drift: Data was collected between April and November 2022. Platform usage patterns and the social media landscape have shifted since then particularly the growth of TikTok which may limit the generalisability of the model to current populations.
- Binary target simplification: Collapsing four risk tiers into two classes loses granularity. A multiclass classification approach (e.g., ordinal logistic regression) would preserve the original tier structure and may offer richer findings.

8. CONCLUSION

This paper presented a logistic regression model for classifying social media users into High or Low mental health risk categories based on self-reported behavioural survey items. Trained and evaluated on 481 respondents, the model achieves strong performance across accuracy, F1 score, and ROC-AUC metrics, with cross-validation confirming that results are stable and generalisable rather than the product of a favourable single split.

The feature importance analysis identifies `Bothered_By_Worries`, `Depression_Frequency`, `Restlessness`, `Sleep_Issues`, and `Purposeless_Usage` as the strongest predictors of High Risk classification and daily screen time as a comparatively weaker one. This finding, consistent across both the Milestone 4 regression analysis and the current classification model, suggests that future research and intervention efforts should prioritise the quality and compulsiveness of social media engagement over raw usage duration.

Future extensions of this work could include multiclass ordinal logistic regression to preserve the full four-tier risk structure, ensemble methods such as Random Forest or Gradient Boosting for improved accuracy, longitudinal data collection to enable causal modelling, and the integration of platform-specific usage data to assess whether different platforms carry differential risk profiles.